

HW08

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Exercise 1

Collaborators: Sam

Tools and resources used

Exercise 2

- Points attempted: 1
- Self-assessed score: 1/1.
- Time spent: 2 minutes.
- Difficulty: 1/10.
- Questions/Feedback: None..

HW8 Exercise 2: One thing that I would like to see on HW 9 would be some more work on burnside's theorem. Although I "know" the theorem, I do not have a full intuition for it yet.

Exercise 3

- Points attempted: 10
- Self-assessed score: 10/10.
- Time spent: 100 minutes.
- Difficulty: 8/10/10.
- Questions/Feedback: Correctness, clarity.

1 Topology up to the Fundamental Group, and why we care

(work in progress title)

Many of us have probably heard something along the lines of "topologist view donuts and coffee mugs as the same thing", but I feel like that is a very poor representation of what the field of topology has to offer. Topology is one of the most intriguing and beautiful branches of math, that has some truly deep foundational insights on computer science, engineering, and the very *structure* of mathematics itself, and this view of topology is really a disservice. In this article, I intend to provide you, the reader, with a more complete view of topology, and why we even care.

1.1 The building blocks of topology

Now, the "real" foundations of topology lie in something known as metric spaces¹. However, I feel as though we can still form a complete enough view of topology and its uses if we accept some abstractions. A true deep dive into metric spaces touches on topics like continuity, and even Real Analysis, which is a bit out of scope for us. Despite this, we still need a some grasp on what a metric space is.

1.1.1 Metric Spaces

Simply put, a metric space is a set of points, such that we have some notion of distances (there are some special requirements!) between them, ie. a **metric**. Formally, we have a set of points X and a $d(x_1, x_2)$ is a metric. Together, (X, d) form a **metric space**.

For (X, d) to be a metric space, d must satisfies some constraints. Namely:

1. d must take in two points of $x_1, x_2 \in X$, and give us a real number $r \geq 0$. Formally, $d \colon X \times X \rightarrow [0, \infty)$.
2. d must be symmetric, so $d(x_1, x_2) = d(x_2, x_1)$.
3. $d(x_1, x_2) = 0$ if and only if $x_1 = x_2$.
4. d satisfies the **triangle inequality**. So for, $x, y, z \in X$, $d(x, z) \leq d(x, y) + d(y, z)$.

Notice that we actually haven't said anything about the set X , and that is because we actually don't need to, if we are strictly talking about metric spaces. For example, for those in computer science, we may recognize these requirements as the exact same requirements needed for a heuristic function for path finding algorithms like A*!

However, for topology specifically, we do need the set X to be **continuous**. There is a very formal and mathematically rigorous definition for continuity², but we only need a sort of intuition for what it means. I am sure your imagination won't lead you astray here.

¹<https://ncatlab.org/nlab/show/metric+space>

²<https://ncatlab.org/nlab/show/topology#Continuity>

1.1.2 Open Sets

One idea that is crucial to the understanding of topology is the notion of the **open set**. Be prepared, as this is one of the most mind bending concepts we have to visit.

Before we get into any formal definitions, I would like to invite you to consider an example. Suppose we have the set S of real numbers $0 < s < 1$ (the exclusive range is very important). Can we define a continuous, bijective mapping f that takes S and to the range $(0, \infty)$?

S is actually an example of an open set. I will explain why in the next section.

Yes! We can say $f(x) = -\log(x)$.

Now, what if we let our input be the set Q with values in the *inclusive* range $[0, 1]$, can we define a similar mapping to the to $[0, \infty)$?

Surprisingly, no! If we take the same function $f(x) = -\log(x)$, this works for the "left" side of the desired input range since $f(1) = 0$. However, f is undefined for $x = 0$. Somehow, *expanding* the input range *shrinks* the output range!

If you don't believe me, try it yourself! There is no continuous bijective function that satisfies these requirements. You can construct a discontinuous function, but constructing a continuous function is impossible.

Now, this is where I start to get excited, because in topology, we often speak about this thing called "continuous deformations", where we arbitrary stretch a shape without tearing or gluing. For example, the thing where we morph a donut into a coffee mug and vice versa. Intuitively, we can kind of see how we can stretch the open set $(0, 1)$ into the open set $(0, \infty)$, but we can't do the same for $[0, 1]$. This our first taste of stretching things around!

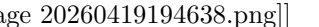
Ok, I hope that you can begin to see why open sets are important to our discussion about donuts and coffee mugs, but if not, I hope that you at least thought it was interesting how adding numbers to our range can actually clamp the range of the output.

1.1.3 Open sets, formally

Open sets can only be defined on continuous metric spaces. Yes, that means that the range $S = (0, 1)$ is a continuous metric space.

I invite you to define as many metric functions d as you can. Here are some examples to get your mind going:

1. $d(x, y) = |x - y|$
2. $d(x, y) = (x - y)^2$

Now, let us pick a point $x \in S$. We a **neighborhood** $N \subset S$ around x is defined as some set of points around x . Within our neighborhood N , there exists an **open ball** B with radius ϵ . In other words, $B = \{y \mid d(x, y) < \epsilon\}$. 

All of that is a fancy way of saying that if we have a point x , we can wiggle it around a bit by ϵ and we are still in S .

Now, for a set to be **open**, every point $x \in S$ must have some neighborhood around it. For example, let us take a look at our example from earlier, $S = (0, 1)$.

Obvious, a value like 0.5 will have a neighborhood around it, and a pretty big neighborhood at that. However, what about values like 0.001 ? Sure they do! Just wiggle it around by 0.0001 ! As a matter of fact, no matter what value we pick, we can *always* create a neighborhood around it, so S is open. Try it yourself! 

This openness (the ability to wiggle things around) is the precise reason why our mapping f could be constructed from $(0, 1)$ to $(0, \infty)$, but not from $[0, 1]$. (TODO, need some more intuition behind this)

1.1.4 Topological spaces

Alright, We have done a *lot* of "tool-making". But now, all of the mathematical machinery is in place for us to begin tackling topological spaces! In other words, we are ready to start talking about proverbial donuts and coffee mugs.

Choice Exercise 5

- Points attempted: 10
- Self-assessed score: 10/10.
- Time spent: 30 minutes.
- Difficulty: 5/10.
- Questions/Feedback: None..

We defined a (left) group action of the group $(G, *)$ on the set X as a function $\cdot : G \times X \rightarrow X$ such that for all $x \in X$ and $g_1, g_2 \in G$

- $e \cdot x = x$, and
- $g_1 \cdot (g_2 \cdot x) = (g_1 * g_2) \cdot x$.

1. [4] For each of the following subparts, give an example of a group $(G, *)$, a set X , and a function $\cdot : G \times X \rightarrow X$ such that

(a) The function is a group action (i.e. it satisfies both of the above properties.)

We can have the group $SO(3)$ acting on the set of 3D vectors $\mathbb{R}^3$, where the function is a matrix-vector multiplication. The identity matrix $I \in SO(3)$ satisfies the first condition, and the associativity of matrix-vector products satisfy the second condition.

(b) The function satisfies the first property, but not the second.

We can have the group C_2 and the set $\mathbb{Z}$, with the function being addition, such that the identity in C_2 corresponds to 0 and r corresponds to 1. Now, the first condition is always met, however the second condition breaks.

(c) The function satisfies the second property, but not the first.

Now, we can have the group C_2 and the set $\mathbb{Z}/2\mathbb{Z}$, with the function

$$c \cdot x = \begin{cases} 1 + x & c = e \\ x & c = r \end{cases},$$

for $c \in C_2$ and $x \in \mathbb{Z}/2\mathbb{Z}$.

(d) We can The function satisfies neither property.

We can have any non-trivial group G on any set X such that $|X| > 1$, where the function always maps to some $x \in X$. This function breaks both conditions.

2. [3] Prove that if G is a group that acts on a set X , then the set

$$H = \{g \in G \mid g \cdot x = x \text{ for all } x \in X\}$$

is a normal subgroup of G .

Proof. To begin, we note that for H to be a normal subgroup of G , H itself must be a group. So, we must show that H satisfies the group axioms. First, associativity is inherited. Also notice that, by definition, $e \in H$.

Then, H is closed because for some $h, h' \in H$, $h \cdot x = x$ and $h' \cdot x = x$, so $h \cdot (h' \cdot x) = x$. This means that $(h * h') \cdot x = x$, so $h * h' \in H$. Thus, H is closed.

Now, we must prove that for all $h \in H$, $h^{-1} \in H$. First, we know that $e \in H$, because $e \cdot x = x$. Notice that $h^{-1} * h = e$, so $(h^{-1} * h) \cdot x = x$. By definition of a group action, $(h^{-1} * h) \cdot x = h^{-1} \cdot h \cdot x = h^{-1} \cdot x = x$. Thus, $h^{-1} \in H$.

Now, to show that $H \triangleleft G$, we must show that for any h, g , $g * h * g^{-1} \in H$. Let us directly apply this expression onto some $x \in X$ to get $g \cdot h \cdot g^{-1} \cdot x$. Notice that $g^{-1} \cdot x \in X$, and $h \cdot x' = x'$ for any $x' \in X$. So, we can effectively remove h from the expression to get $g \cdot g^{-1} \cdot x = (g * g^{-1}) \cdot x = x$. Thus, $g * h * g^{-1} \in H$, so $H \triangleleft G$. $\square$

3. [3] Let $G = \mathbb{Z}$, $X = (H, *)$, and $h \in H$ be an element of H . Prove that the map $n \cdot h = h^n * h * h^{-n}$ is a group action.

Proof. First, we prove that $e \cdot x = x$. Let $e \in G = 0$. Now, $e \cdot h = h^0 * h * h^0 = h$. This satisfies the first condition.

To prove the second condition, let $n, n' \in G$, so

$$n \cdot (n' \cdot h) = n \cdot (h^{n'} * h * h^{-n'}) \quad (1)$$

$$= h^n * h^{n'} * h * h^{-n'} * h^{-n} \quad (2)$$

$$= h^{n+n'} * h * h^{-(n+n')} \quad (3)$$

$$= (n + n') \cdot h. \quad (4)$$

Therefore, the map $n \cdot h = h^n * h * h^{-n}$ is a group action. $\square$

Choice Exercise 7

1. [3] Set a timer for 25 minutes and read an algebra text of your choosing, attempting problems/exercises along the way.]

I chose to read chapter 6, in part 3 of the Infinitely Large Napkin, which covers the concept of the metric space. I was able to do 3 exercises in total.

Exercise 6.1.3. Use the triangle inequality to show these are equivalent. (The names “radius” and “diameter” are a big hint!)

A metric space $M = (M, d)$ is bounded if there is a constant D such that $d(p, q) \leq D$ for all $p, q \in M$.

$$d(p, r) \leq d(p, q) + d(q, r) \quad (5)$$

$$d(p, q) \leq R, d(q, r) \leq R \quad (6)$$

$$d(p, r) \leq 2R = D. \quad (7)$$

Exercise 6.1.6. Show that “totally bounded” implies “bounded”.

If a space is totally bounded, then it can be covered by finitely many ε -neighborhoods, for $\varepsilon > 0$.

Therefore, in the worst case, our diameter D will be $n\varepsilon$, where n is the number of ε -neighborhoods.

Question 6.2.2. Show that a sequence which converges is automatically Cauchy.

A sequence $x_1, x_1, \dots$ in a metric space M is Cauchy if for any $\varepsilon > 0$, $d_M(x_m, x_n) < \varepsilon$, for all sufficiently large m and n .

I will borrow an old definition of converges from Intro to Proofs:

A sequence $y_1, y_2, \dots \in \mathbb{R}$ converges to c if

$$\forall \varepsilon > 0, \exists m \in \mathbb{N}, \forall n > m, |y_n - c| < \varepsilon.$$

We have a couple moving peices here. In our defintion for convergence, we can view $\mathbb{R}$ as a metric space, where $d(p, q) = |p - q|$. Also, this definition only tells us the distance between our sequence at index m and some constant c , not the distance between two items y_n, y_m . However, by the triangle inequality, we can see that, given $d(y_n, c) < \varepsilon$ and $d(y_{n'}, c) < \varepsilon$ for $n' > n$, $d(y_n, y_{n'}) < d(y_n, c) + d(y_{n'}, c) = 2\varepsilon$. This satisfies our definition of Cauchy.